

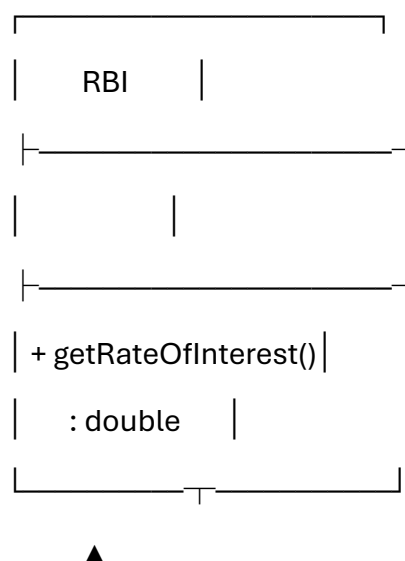
Aim

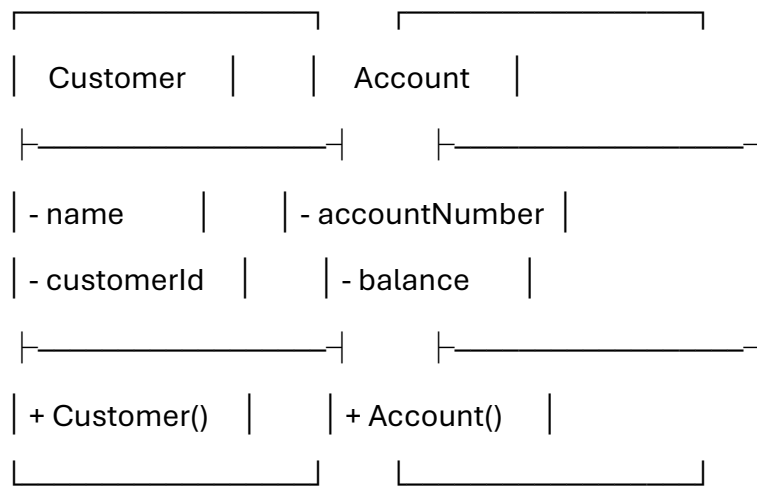
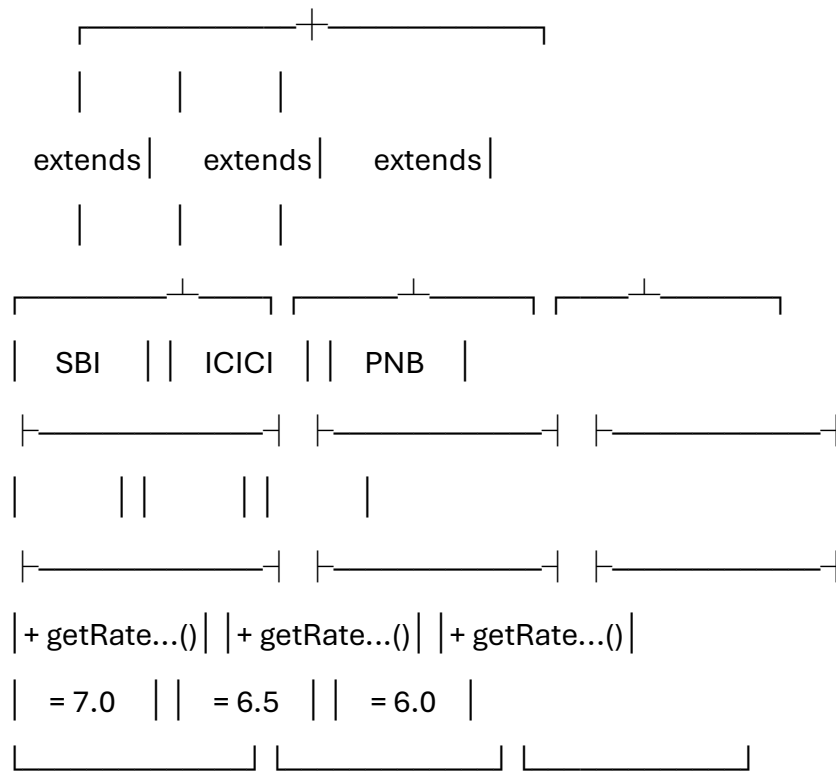
To write and execute a Java program to demonstrate **method overriding and dynamic polymorphism** using RBI as the base class and SBI, ICICI, and PNB as derived classes.

Algorithm

1. Start the program.
2. Create a Customer class with customer details.
3. Create an Account class with account details.
4. Create an RBI base class with a method `getRateOfInterest()`.
5. Set the minimum RBI interest rate as **4%**.
6. Create derived classes SBI, ICICI, and PNB.
7. Override the `getRateOfInterest()` method in each derived class.
8. Create an RBI reference.
9. Based on the bank name entered by the user, assign the corresponding bank object to the RBI reference.
10. Call `getRateOfInterest()` using the RBI reference.
11. The overridden method of the actual object is executed, demonstrating **dynamic binding**.
12. Display the rate of interest.
13. Stop the program.

UML Diagram :





Code:

```
package Records;
```

```
import java.util.Scanner;
```

```
// Customer class

class Customer {

    String name;

    int customerId;


    Customer(String name, int customerId) {

        this.name = name;

        this.customerId = customerId;

    }

}
```

```
// Account class

class Account {

    int accountNumber;

    double balance;


    Account(int accountNumber, double balance) {

        this.accountNumber = accountNumber;

        this.balance = balance;

    }

}
```

```
// RBI Base Class

class RBI {


    double getRateOfInterest() {

        return 4.0;

    }

}
```

```
}
```

```
// SBI Derived Class
```

```
class SBI extends RBI {
```

```
    @Override
```

```
    double getRateOfInterest() {
```

```
        return 7.0;
```

```
    }
```

```
}
```

```
// ICICI Derived Class
```

```
class ICICI extends RBI {
```

```
    @Override
```

```
    double getRateOfInterest() {
```

```
        return 6.5;
```

```
    }
```

```
}
```

```
// PNB Derived Class
```

```
class PNB extends RBI {
```

```
    @Override
```

```
    double getRateOfInterest() {
```

```
        return 6.0;
```

```
    }
```

```
}
```

```
// Main Class
```

```
public class BankInterest {
```

```
    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        System.out.print("Enter the Bank name to find the rate of Interest : ");
```

```
        String bankName = sc.nextLine();
```

```
        // Dynamic polymorphism
```

```
        RBI bank;
```

```
        if (bankName.equalsIgnoreCase("RBI")) {
```

```
            bank = new RBI();
```

```
            System.out.println("RBI rate of interest is : "
```

```
                + bank.getRateOfInterest() + "%");
```

```
        } else if (bankName.equalsIgnoreCase("SBI")) {
```

```
            bank = new SBI();
```

```
            System.out.println("RBI rate of interest is : "
```

```
                + bank.getRateOfInterest() + "%");
```

```
        } else if (bankName.equalsIgnoreCase("ICICI")) {
```

```
            bank = new ICICI();
```

```
            System.out.println("RBI rate of interest is : "
```

```
                + bank.getRateOfInterest() + "%");
```

```

    } else if (bankName.equalsIgnoreCase("PNB")) {
        bank = new PNB();
        System.out.println("RBI rate of interest is : "
            + bank.getRateOfInterest() + "%");

    } else {
        System.out.println("Invalid Bank Name");
    }

    sc.close();
}
}

```

Output:

```

Records.BankInterest'
Enter the Bank name to find the rate of Interest : sbi
RBI rate of interest is : 7.0%
PS C:\Users\Sahithi_pyla\Desktop\Sahithi_java>

```